

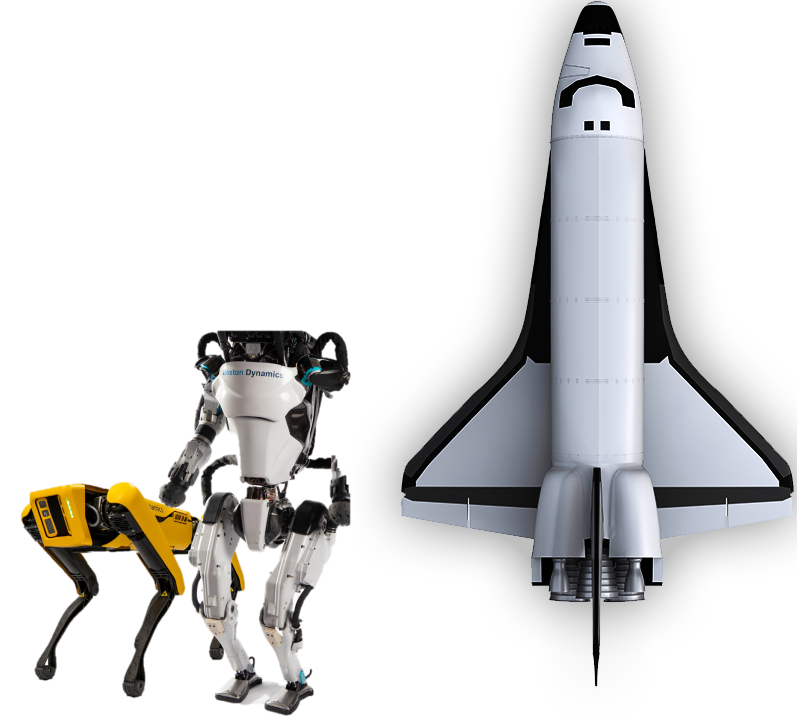
EEEE4064

# ADVANCED CONTROL SYSTEM DESIGN

## Lecture 1 Introduction



University of  
Nottingham  
UK | CHINA | MALAYSIA



# Module Information

## TARGETED STUDENTS

- MEng and MSc students of Electrical and Electronic Engineering

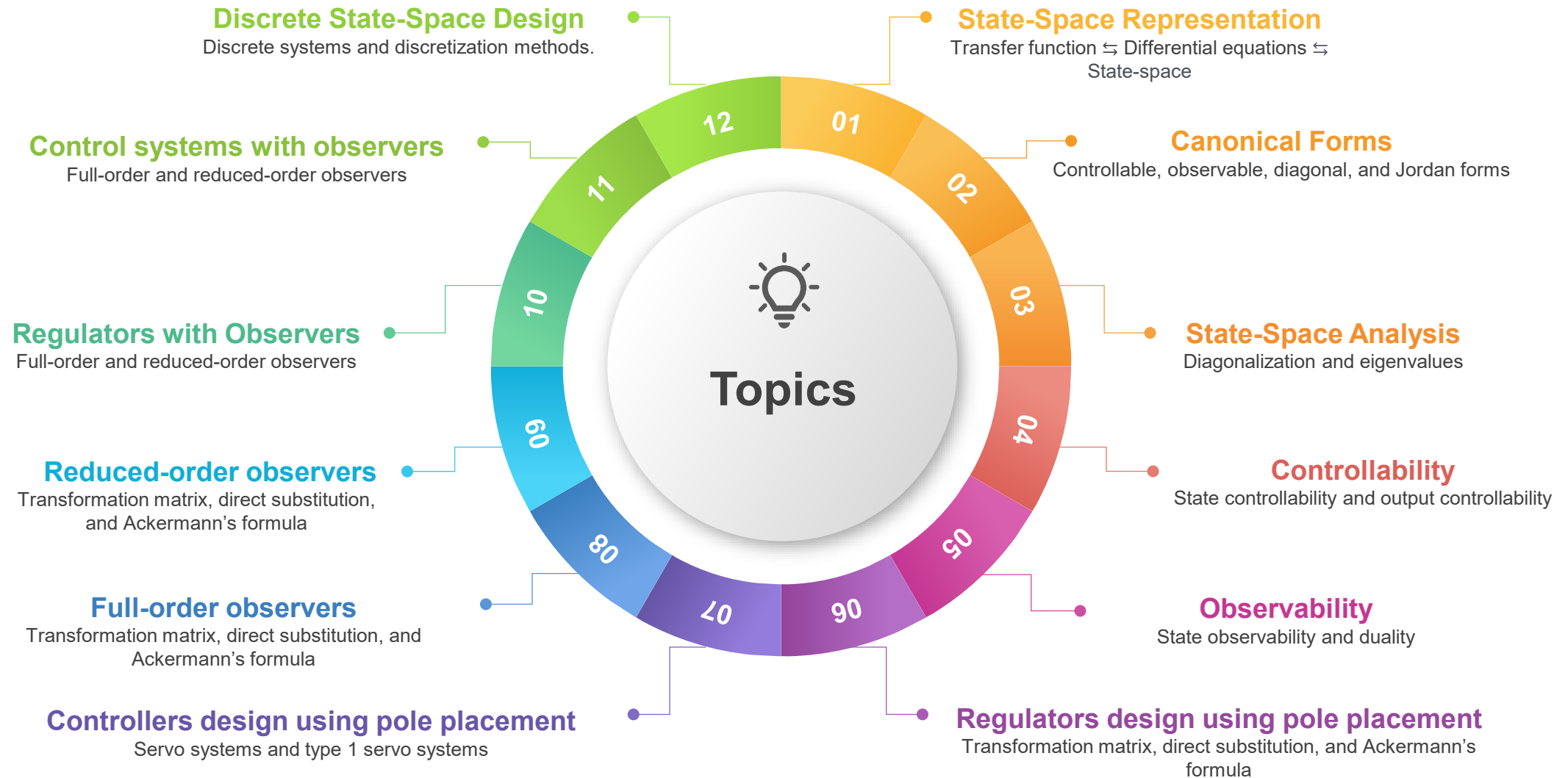
## LEARNING OUTCOMES

1. Apply multivariable state space techniques to determine models of linear systems.
2. Analyze state space models and determine controllability and observability.
3. Design multivariable state feedback control systems in continuous and discrete domains.
4. Design full order and reduced order observers for state variable estimation in continuous and discrete domains.

## ASSESSMENTS

- 20% Coursework 1. Mainly on system modeling and analysis.
- 20% Coursework 2. Mainly on control design and analysis.
- 60% Final Exam.

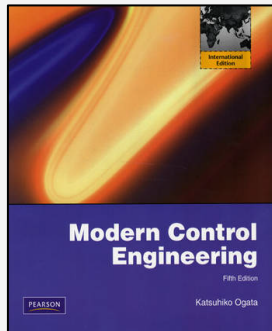
# Topics



# Textbooks

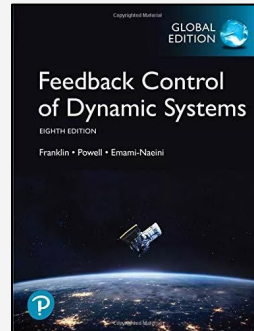
01

K. Ogata, Modern  
Control Engineering,  
5th Ed., 2010,  
Pearson.



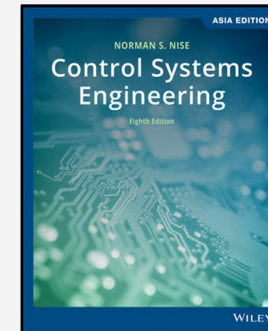
02

F. Franklin, et al.,  
Feedback Control of  
Dynamic Systems, 8th  
Ed., 2020, Pearson.



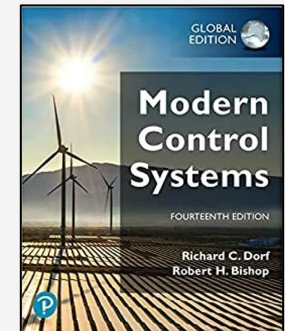
03

N. Nise, Control  
Systems Engineering  
8th Ed., 2019, Wiley.



04

R. Dorf, and R. Bishop,  
Modern Control  
Systems 14th Ed.,  
2022, Pearson.



# Classical and Modern Control

## Classical Domain

Deals with the behavior of dynamical systems with inputs, and how their behavior is modified by feedback, using the **Laplace transform** as a basic tool to model such systems.

## Classical Abilities

Deals with linear time-invariant (LTI) single-input single-output (**SISO**) systems. The SISO relationship can be represented using a **transfer function** that relates the Laplace transform of the output to the input.

## Modern Domain

Converts the differential equations of a system into a system of lower-order time-domain equations called **state-space equations**.

## Modern Abilities

The state-space equations be manipulated using techniques from linear algebra. This allows dealing with multi-input multi-output (**MIMO**) systems using **state equations** and an **output equations**.



# Classical and Modern Control

- Transfer function
- s-domain
- s-plane
- Root locus
- Bode plot

## CLASSICAL CONTROL

$$G(s) = \frac{1}{s^3 + 2s^2 + 5s + 8}$$

- State-space
- Time-domain
- Matrix operations
- Linearization
- Pole placement

## MODERN CONTROL

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & -5 & -8 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

# Classical and Modern Control

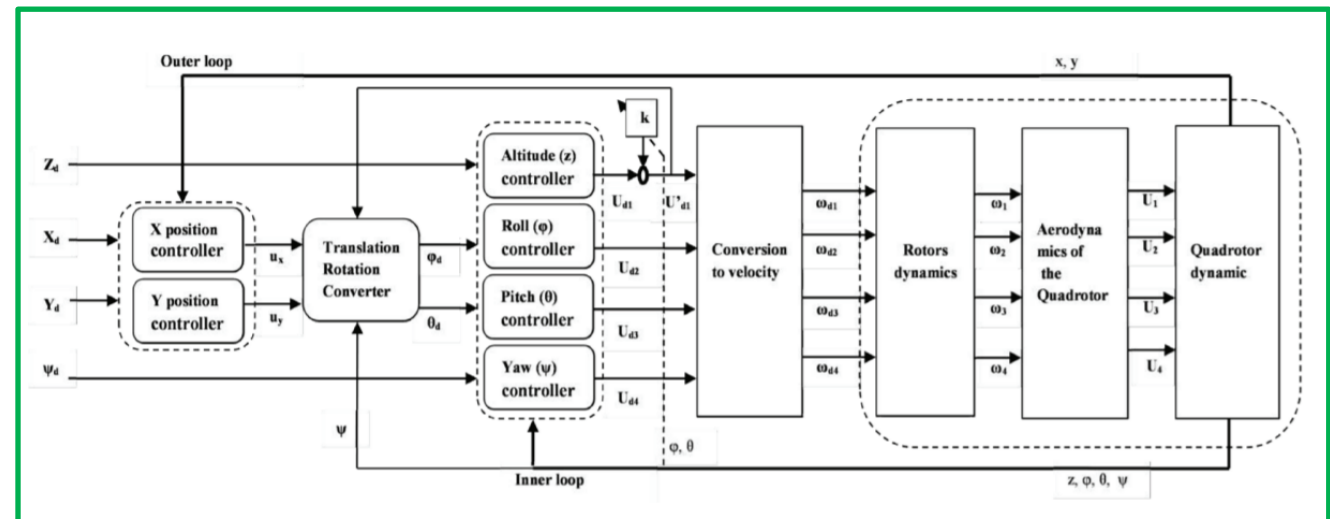
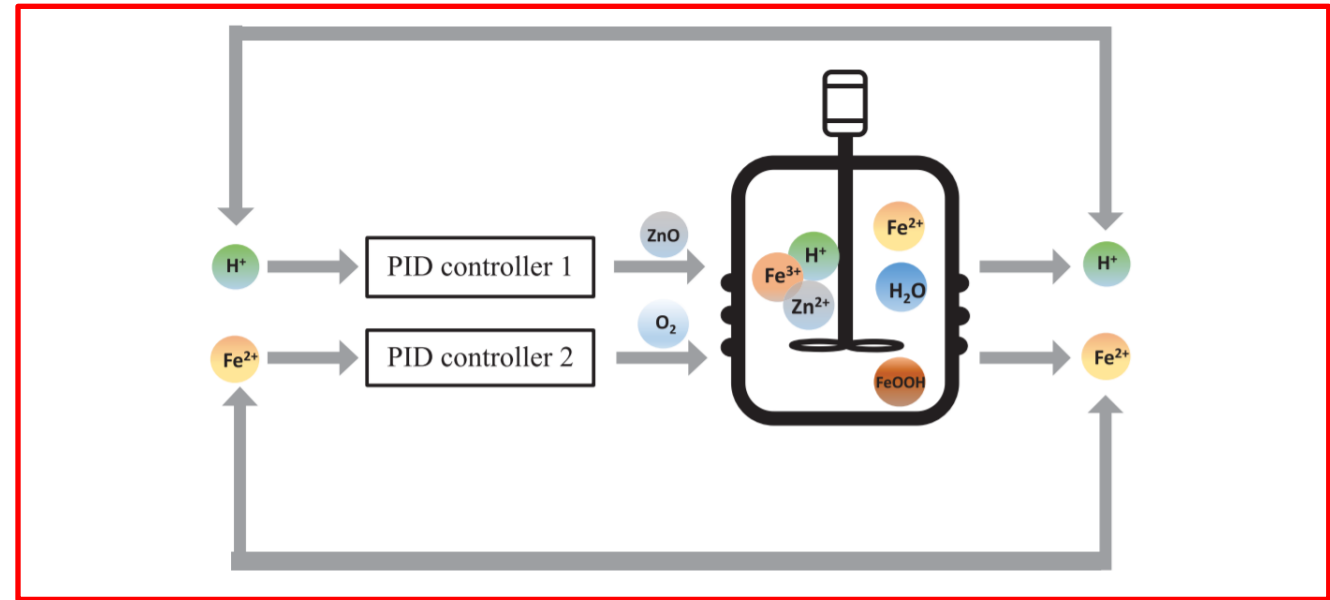
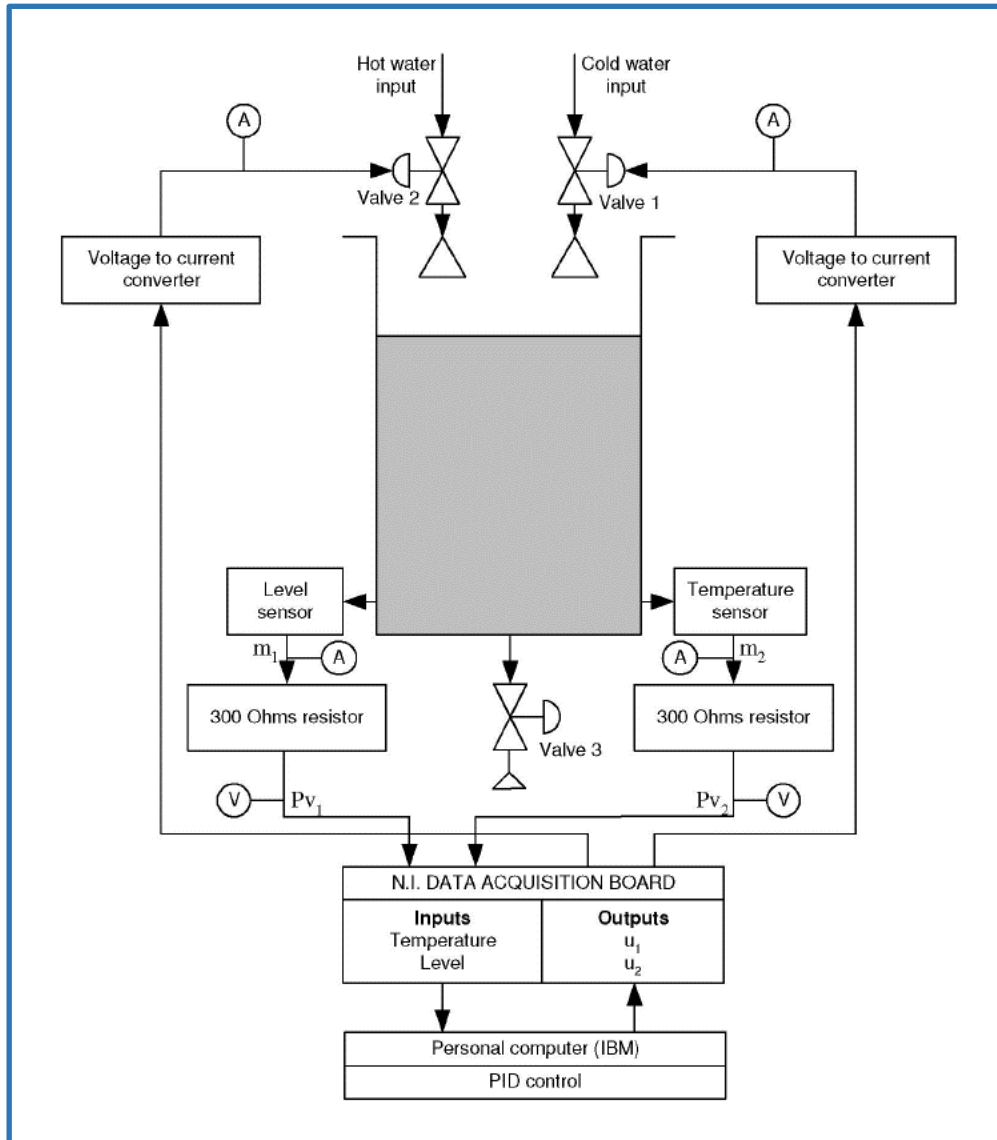
## CLASSICAL

1. Modeling/System Identification
2. Analysis of system impulse response
3. Performance requirements
4. Compensator design
5. Closed-loop system verification
6. Ideal for SISO LTI systems
7. Easier for humans to solve

## MODERN

1. Modeling/System Identification
2. Convert to state-space
3. Analysis of system impulse response
4. Performance requirements
5. Compensator design
6. Closed-loop system verification
7. Can handle MIMO systems well
8. Easier for computers to solve

# Case Studies



# State-Space

01

The third control design technique after **root-locus** and **frequency response**.

02

Operations are done in **time domain**, using **differential equations**.

03

Defines system **state variables** instead of transfer function.

04

Complex systems can be analyzed more easily after linearization to produce a **vector-matrix** consisting of  $n$  first order differential equations.

05

Able to handle **MIMO** systems directly.

06

Computational operations can be easily **programmed**.

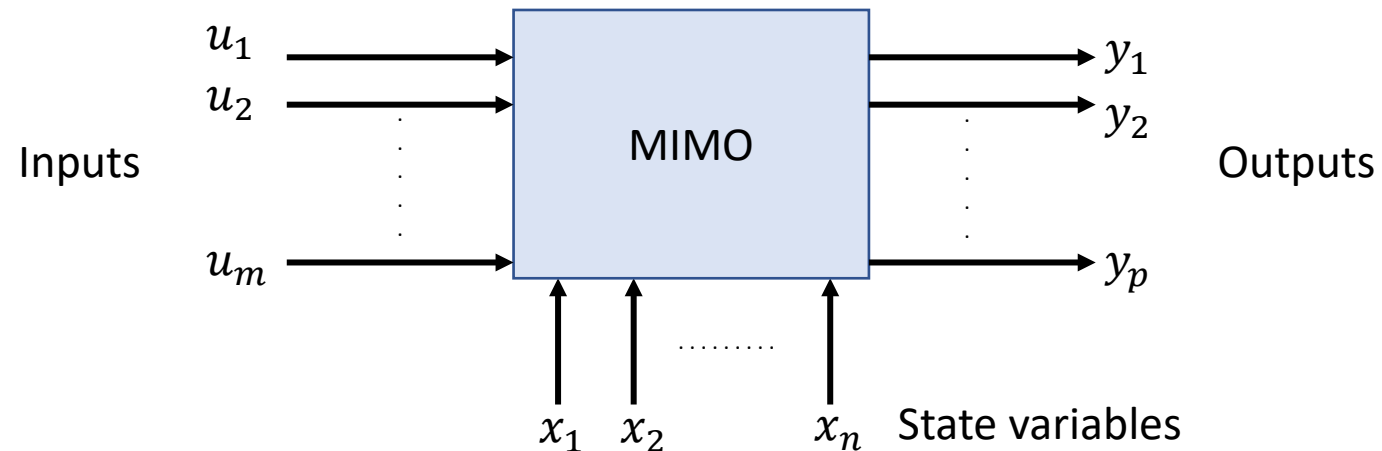
# State-Space

07

The **state variables** of a dynamic system are the variables making up the smallest set of variables that determine the state of the dynamic system.

08

A **state-space representation** is a mathematical model of a physical system as a set of input, output and state variables related by first-order differential equations or difference equations.



# *State-Space Representation*

*Next Week ...*

**DO YOU  
HAVE ANY  
QUESTIONS**